Lifting dfranzen's ARC-AGI-3 harness: what moved the hidden-game score

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Abstract

We tested 21 changes on top of dfranzen's Milestone 2 notebook for the ARC Prize 2026 ARC-AGI-3 Kaggle competition (offline, 9 hours, one RTX PRO 6000, 110 hidden interactive games, RHAE score). Changes were screened on a 25-game local run, on a 40-episode stuck-level suite, and by public submission. One configuration, the D′ slot priority from a public notebook, leads plain dfranzen by 1.7 public points over three draws each ($p \approx 0.34$). No change we built moved the public score, and no local or suite result outside the noise band reproduced.

1 Setup

The base is dfranzen's notebook: Qwen3.8-Flash-Next (W4A16) with a multi-token-prediction drafter, served by SGLang "Pennyroyal" 2.5.3, driving a Python-tool agent harness over 10 concurrent game streams with a priority scheduler. Kaggle allows one submission per day; the public score covers about 55 of the 110 hidden games.

Three screens, with their measured noise:

- **Public score**: identical plain copies scored 28.39, 27.86 and 24.60 (sd 2.05); community copies average 25.77 (sd 3.93).
- **Local 25-game run** (RHAE on the public games): identical runs differ by up to about 7 points.
- **Stuck-level suite**: 20 levels where full runs got stuck, each replayed twice from the recorded history; identical baselines cleared 9 and 9 of 40 episodes (sd about 3.7), so a gain needs about 16.

2 Funnel

Stage	Count
Mechanisms built	21
Tested on the 25-game local run	17
Above the local noise band (42.9 ± 7)	1
Reproduced when stacked or re-run	0
Submitted publicly (as changes on dfranzen)	4
Beat plain dfranzen's public mean	1
Significant against plain ($p < 0.05$)	0

3 Measurements

Public submissions

Date	Configuration	Public
10-02	dfranzen + level prior + fade	20.60
10-03, 10-04, 10-05	plain dfranzen copy	28.39, 27.86, 24.60
10-06, 10-07, 10-09	D′ slot priority (shiiin9)	29.68, 29.71, 26.60
10-08	dfranzen + endgame FIFO rotation	22.56
10-10	D′ + re-attempt, new kernel slug	0.00 (rerun failed)
10-11	D′ + re-attempt, proven kernel	26.05

Local 25-game runs

Change	Score	Levels	tok/s
unmodified (mean of 3)	42.9	101	597
re-attempt v4 (keeps slot)	50.18	109	578
HiCache, 160K context	48.63	108	530
level prior + fade (2 runs)	48.6	104	–
endgame FIFO rotation	47.88	109	–
HiCache 160K + re-attempt	42.56	98	547
256 px board images	42.70	99	627
re-attempt v3 (lost its slot)	42.31	95	542
24K history drain	44.93	97	416
reasoning window	24.27	74	544

Stuck-level suite (episodes cleared of 40; baseline 9, 9)

Stall probe 10; prompt candidates 11, 11, 12, 9, 5; win records 8; thinking cap 8; keep-context restart 8.

Re-attempt conversion

Of 28 re-attempts (v3 and v4), 6 cleared their level afterwards (21%), against about 25% of stuck levels cleared by continuing in unmodified runs.

4 Findings

1. Scheduler changes that move time between games won locally and lost publicly (20.60, 22.56): the 25-game local mix does not represent the 110 hidden games.
2. The model depends on its own past reasoning: removing it collapsed the run to 24.27.
3. Restarting a stuck level, with or without its context, converts it no more often than continuing.
4. Smaller board images (256 px) cost nothing and add 5% throughput.
5. A submission from a newly created kernel scored 0.00 although its save run passed; the same code from an existing kernel scored normally.

5 What is not established

- Whether D′ beats plain dfranzen: resolving 2 points needs about 60 draws per arm.
- Any local effect under about 5 points.
- The cause of the new-kernel 0.00: no scored-rerun logs are available; a Python 3.13 image change, a dataset-version race and mount failures all fit.
- What the teams above 40 do: none of the top five has disclosed a method.
- Whether 256 px images change the public score.

Sources

Results: `docs/experiments/arc-agi3-ledger.json`. Dashboard: `https://claude.ai/artifact/LsbFyTwgMn4EmDU2vN8mko`. Literature: `reports/ARC AGI 3 SOTA October update.md`.